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A demonstration of a digital twin framework for structural health monitoring: Application to bridge infrastructures

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ABSTRACT

This study introduces a novel, multi-layered Digital Twin (DT) framework designed to enhance the resilience of ageing bridge infrastructure through real-time structural health monitoring (SHM) and data-driven decision support. The proposed framework integrates physics-based Finite Element Modelling (FEM), drone-based photogrammetry, and wireless sensor networks to construct a dynamic digital representation of the physical asset. By continuously synchronising sensor data with virtual models, the system establishes a foundation for predictive maintenance and lifecycle optimisation. Key innovations include a modular architecture that supports the seamless integration of diverse data sources, a closed-loop feedback mechanism for iterative model updating, and functionality for real-time anomaly detection. The proposed system supports proactive monitoring by enabling dynamic condition tracking, structural behaviour analysis, and long-term trend forecasting. The framework has been demonstrated on an operational railway truss bridge, where live vibration and environmental data were used to calibrate and validate the DT in a real-world setting. The results underscore the system's potential as a robust and scalable monitoring solution for historically significant and ageing transport assets. This work addresses critical limitations of conventional SHM approaches by offering a unified, data-centric strategy for infrastructure management. Beyond operational awareness, the proposed DT platform provides a strategic pathway toward more intelligent, more sustainable infrastructure systems prioritising resilience, informed maintenance planning, and future adaptability.

1. Introduction

Bridge infrastructures are vital assets that ensure connectivity, support economic activity, and build societal resilience. However, ageing structures face rising risks from degradation, environmental stress, and heavy loads. These risks can harm safety and lead to expensive repairs or failures. Overcoming these challenges requires advanced methods that allow continuous monitoring, predictive maintenance, and efficient resource allocation.

Traditional Structural Health Monitoring (SHM) methods, such as periodic visual inspections and manual data collection, have provided valuable insights, yet they remain limited by subjectivity, lack of real-time capabilities, and an incomplete characterisation of dynamic structural behaviour (Ye et al., 2012). Similarly, finite element modelling (FEM) depends heavily on assumptions about material properties and boundary conditions and requires frequent calibration to remain accurate (Gomez-Cabrera and Escamilla-Ambrosio, 2022), (Liu et al., 2009). These limitations underscore the need for integrated approaches that combine continuous sensing with adaptive modelling.

Digital Twin (DT) technology represents a paradigm shift in infrastructure management by creating synchronised virtual counterparts of physical assets. Through the integration of real-time sensor data, numerical simulations, and advanced analytics, DTs enable predictive maintenance, early anomaly detection, and intuitive visualisation of structural performance (Wang et al., 2022), (Armijo and Zamora-Sánchez, 2024), (Ahmed et al., 2024). In SHM, this evolution has shifted the focus from damage detection toward proactive, data-driven strategies (Bao and Li, 2020).

Despite significant progress in sectors such as manufacturing and aerospace, DT applications in civil infrastructure remain limited. Existing frameworks often focus on isolated functions, such as geometric reconstruction using drones (Zhou et al., 2021), or anomaly detection

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via machine learning, rather than achieving full integration of FEM, real-time sensing, and predictive analytics. Comprehensive hybrid frameworks that combine physics-based and data-driven approaches with operational field validation are still lacking.

This study addresses that gap by presenting a hybrid DT framework for bridge SHM, coupling physics-based FEM with IoT-enabled sensing in a continuous feedback loop. The proposed system couples FEM, drone-based photogrammetry, and sensor networks into a real-time, closed-loop feedback mechanism. Signal processing techniques (Fast Fourier Transform (FFT) and Frequency Domain Decomposition (FDD)), data assimilation methods (e.g., Kalman filtering, Bayesian updating), and threshold-based event detection ensure that the system continuously calibrates to observed structural behaviour (Ye et al., 2019a), (Liu et al., 2023a).

The research investigates three core aspects: (1) it examines the design and integration of the framework, combining physics-based modelling, and IoT-enabled sensing into a unified DT–SHM platform, (2) the reliability of the framework for event and condition detection, and (3) the scalability of the approach across diverse bridge typologies, including concrete, steel, and timber bridges, as well as complex configurations such as suspension bridges and other ageing infrastructure contexts.

The novelty of this work lies in its full-scale implementation and field validation on operational road and rail bridges. Unlike existing DT-SHM approaches that often remain confined to a single domain in the research phase, such as FEM, geometric reconstruction, or machine learning, the proposed practical framework demonstrates a fully integrated system on a real bridge. The framework is capable of continuously integrating multi-source data streams, synchronising them with physics-based models, and analysing structural behaviour in near real-time. Beyond serving as a technical demonstrator, the system is positioned as an innovative asset management tool, enabling proactive decision-making by combining monitoring, analysis, and predictive insights within a unified digital twin environment. The main contributions of this paper are:

1. A hybrid modelling platform that integrates FEM and data-driven models through robust signal processing and model updating strategies.
2. A scalable data pipeline that ensures continuous DT synchronisation, enabling timely anomaly detection and proactive maintenance.
3. Empirical demonstration of the framework on an operational railway truss bridge, validating its adaptability for ageing and deteriorated assets.
4. Critical reflection on the limitations of real-time monitoring and scalability challenges across infrastructure typologies.

Overall, this work presents a practical, scalable, and validated DT–SHM integration approach, moving beyond theoretical models to offer a data-centric solution for managing ageing infrastructure.

2. Background and literature review

2.1. DT technology in SHM and infrastructure management

Initially, the concept of DT, introduced by Dr Michael Grieves in 2002 at the University of Michigan, and later formalised by NASA, has evolved into a powerful paradigm for representing and managing complex physical systems through their virtual counterparts. A DT integrates real-time sensor data, advanced simulations, and historical information to mirror the behaviour of physical assets under changing conditions (Grieves, 2023), (Glaessgen and Stargel, 2012).

In the context of SHM, DTs enable continuous, synchronised updates between the physical infrastructure and its virtual model. This dynamic interaction supports real-time monitoring, predictive maintenance, and informed decision-making, transforming infrastructure management from reactive to proactive.

While DTs have gained widespread adoption in manufacturing and aerospace, their implementation in civil infrastructure, particularly bridges, remains limited. Many existing frameworks address only isolated components, such as geometric reconstruction via Unmanned Aerial Vehicles (UAVs) imagery or anomaly detection through machine learning, without achieving full integration of sensing, simulation, and analytics in a closed-loop feedback system (Xu and Zhang, 2022).

A few recent studies have demonstrated hybrid approaches. For example, Qin et al. (2023) proposed a physics-data hybrid approach that integrates finite element modelling with real-time measurement data to maintain alignment between a structure and its DT (Qin et al., 2023). Similarly, C. Ye, Butler et al. (2019) integrated BIM, SHM, and geographic information systems (GIS) into a unified infrastructure management platform. Several DT frameworks demonstrate innovative SHM applications integrating AI-driven Operational Modal Analysis (OMA) to enable real-time damage detection. However, these frameworks often stop short of continuous real-time calibration or practical field deployment (Ye et al., 2019b), (García-Macías et al., 2024), (Dang et al., 2022).

Emerging technologies also address challenges in historic and heritage infrastructure. Kong et al. (2023) used photogrammetry and point cloud processing for detailed monitoring of heritage structures (Kong and Hucks, 2023). Torzoni et al. (2024) proposed a predictive DT framework that integrates probabilistic graphical models and deep learning for real-time diagnostics and maintenance planning (Torzoni et al., 2024).

Overall, the integration of DTs with SHM systems has revolutionised infrastructure management. Unlike conventional inspection-based practices, DT platforms provide continuous, real-time insights into structural performance, enabling predictive maintenance that minimises costs, prevents failures, and reduces disruption. SHM systems support this integration by continuously monitoring critical parameters (such as stress, strain, vibration, displacement, velocity, and temperature) and feeding them into DT platforms. This process maintains accurate digital representations of physical structures and enables proactive, data-driven decisions.

2.1.1. Wireless SHM and LoRaWAN for infrastructure monitoring

SHM increasingly relies on wireless technologies to overcome the limitations of traditional wired systems. While wired sensor networks have historically delivered accurate data for modal analysis, their high installation cost, complex maintenance, and lack of scalability make them impractical for widespread deployment, especially on ageing or remote infrastructure.

Wireless systems offer a more flexible and scalable alternative. Among available options, Long Range Wide Area Network (LoRaWAN) stands out for its combination of low power consumption, long-range transmission (up to 10–15 km), and minimal infrastructure requirements. These features make it particularly suitable for bridge monitoring applications where access to power and connectivity is constrained (Loubet et al., 2019).

Compared to other wireless protocols such as Zigbee or 5G, LoRaWAN operates on unlicensed radio bands (e.g., 915 MHz in Australia), enabling long-term, battery-powered operation without continuous connectivity. While 5G offers higher data rates and lower latency, it demands dense infrastructure and power supply. Conversely, LoRaWAN supports low data rates (~0.3–50 kbps) but enables near-real-time event transmission with significantly longer battery life (4–10 years).

In our DT framework, LoRaWAN-enabled sensors played a central role by delivering real-time vibration and environmental data to the digital twin. The network architecture was designed for periodic and event-triggered sampling, with threshold-based alerts for structural anomalies. This configuration ensured energy efficiency while capturing critical dynamic responses such as train crossings.

However, LoRaWAN presented several operational limitations

during the monitoring campaign. Data packet loss was observed in areas with high signal interference, particularly along the freight corridor, resulting in approximately 5 % transmission failures over the first three months. Such losses can disrupt the synchronisation of data streams and diminish the reliability of certain sensor packets within the network. To maintain analytical integrity, uncertain or incomplete packets were excluded from the dataset to prevent propagation of errors in model calibration and event detection. These limitations could potentially be addressed through the implementation of a multi-gateway architecture and adaptive data-retry protocols, which are anticipated to significantly enhance data continuity and lessen the overall impact of packet loss on the digital twin system.

In addition, limited bandwidth restricted the volume of real-time data that could be transmitted, requiring strategic data compression and event-based prioritisation to ensure critical information was delivered promptly. Looking ahead, hybrid approaches that integrate LoRaWAN with edge computing could further improve data processing efficiency and reduce latency, ensuring that essential information is transmitted with minimal delay even under constrained bandwidth conditions.

Moreover, ongoing advancements in energy-harvesting sensors and hybrid communication protocols continue to enhance the reliability and efficiency of LoRaWAN-based SHM systems, making them more robust for future large-scale infrastructure monitoring deployments.

Despite these limitations, LoRaWAN enabled reliable, low-cost, and long-duration deployment across the monitored bridge. Table 1 summarises the key trade-offs between LoRaWAN and other SHM communication protocols, supporting its selection for this application (Polonelli et al., 2018).

2.2. Integration of digital physics-based and data-driven model in SHM

SHM relies on two primary modelling approaches: physics-based simulations and data-driven algorithms. Physics-based models, such as finite element modelling (FEM), simulate structural behaviour under various loading conditions based on mechanics and material properties. While FEM offers detailed insights into stress distributions, modal shapes, and structural response, its accuracy depends on simplified assumptions, such as idealised boundary conditions and material uniformity, which may not hold true in ageing or deteriorated structures (Ye et al., 2019b).

To maintain predictive accuracy, physics-based models must be continuously updated with real-world sensor data. Standard techniques for this data assimilation include the Extended Kalman Filter (EKF), Bayesian updating, and Reduced-Order Modelling (ROM), all of which adjust key FEM parameters (e.g., stiffness, damping) based on discrepancies between measured and simulated responses. These methods enhance model reliability while optimising computational efficiency (Liu et al., 2023a).

The proposed DT framework integrates physics-based modelling with signal processing and statistical analysis for anomaly detection and adaptive calibration. In its current form, the demonstrator emphasises measurement reliability, robust pre-processing (e.g., modal parameter extraction via FFT/FDD), and recursive model updating. Its modular design supports future inclusion of advanced machine learning methods such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs) for pattern recognition and non-linear response modelling, (Flah et al., 2021).

As illustrated in Fig. 1, the hybrid framework operates in a continuous feedback loop. FEM provides baseline modal predictions; these are compared against extracted parameters from field sensors. The residuals inform a real-time calibration process, such as Kalman-based filtering, that updates FEM parameters accordingly. This closed-loop process ensures synchronisation between the digital model and the physical structure, enabling early detection of changes such as stiffness loss, damage initiation, or abnormal behaviour (Liu et al., 2023a), (Liu et al., 2023b).

Importantly, the framework is designed for diverse bridge typologies, including ageing and historically significant, and large-scale systems such as suspension bridges, where uncertainties in material properties and boundary conditions are pronounced. Future enhancements may incorporate Physics-Informed Neural Networks (PINNs), which embed physical laws into neural network training, and automated FEM generation tools to accelerate model setup across large infrastructure portfolios.

By fusing first-principles simulations with data-driven learning, the proposed DT-SHM platform offers a powerful solution for real-time calibration, anomaly detection, and predictive maintenance,

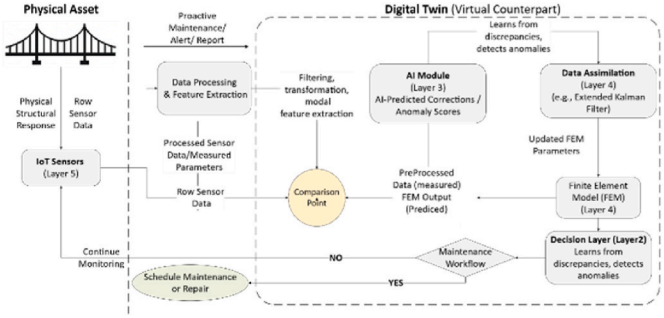


Fig. 1. Schematic of the hybrid AI–FEM calibration loop.

Table 1
LoRaWAN vs. Traditional Wired and Wireless SHM Systems.

Feature	LoRaWAN Wireless	5G Wireless IoT	Zigbee Wireless	Traditional Wired SHM
Installation Cost	Low (no cables, minimal infrastructure)	High (significant infrastructure needed)	Low (short-range networks)	High (expensive cabling and installation)
Maintenance	Minimal (battery-powered, long lifespan 4–8 years)	Moderate (higher power demands 1–3 years)	Moderate (battery replacement needed, 1–2 years)	High (cables and sensors require upkeep)
Scalability	High (multiple sensors per gateway, up to 15 km range)	High (but dependent on infrastructure, up to 10 km range)	Medium (range limitations)	Low (each sensor needs a wired connection)
Data Transmission	Remote, real-time over LoRaWAN and cloud	Ultra-fast (low latency, high data rates)	Short-range, mesh-based	Limited to local data loggers or wired transmission
Data Rate (Typical)	0.3–50 kbps (low)	20–250 kbps (moderate)	20–250 kbps (moderate)	High (Mbps-Gbps)
Latency (Typical)	Seconds to minutes (high)	Hundreds of ms (moderate)	Tens to hundreds of ms (moderate)	Milliseconds (very low)
Energy Per Packet	~10–100 mJ (low, battery lasts 5–10 years)	~100–500 mJ (moderate)	~50–200 mJ (moderate)	Continuous power
Suitability for Bridges	Ideal for long-term, remote monitoring	Feasible but requires reliable connectivity	Feasible for local sensor clustering	Challenging due to wiring constraints
Reliability in Harsh Environments	Moderate (susceptible to interference)	High (dedicated frequency bands)	Moderate (short-range limitations)	Very High (stable and less susceptible to interference)

addressing the limitations of standalone modelling strategies and advancing the state of practice in digital infrastructure management.

2.3. Applications of SHM and DT in aged infrastructure

Aged infrastructure, including early twentieth-century bridges and railways, faces unique challenges due to the deterioration of materials, increased operational loads, and exposure to various environmental stressors. Effective management of these assets requires advanced, data-driven monitoring tools capable of continuous condition assessment, while also considering the need for cultural and historical preservation.

Over time, gradual changes in material properties complicate the assessment of aged infrastructure (Nasim and Setunge, 2024). For example, reinforced concrete is highly susceptible to corrosion, which affects its long-term performance (Casprini et al., 2022), while masonry structures undergo slow but irreversible deterioration (Vicente et al., 2017). Climate change and natural hazards further exacerbate these issues, with fluctuating flood risks altering asset vulnerability profiles (Nasim and Setunge, 2022). DT platforms equipped with SHM capabilities can simulate such scenarios, evaluating how ageing materials and environmental conditions influence structural behaviour (Kioumarsi et al., 2022).

Traditionally, SHM for aged structures relied on manual inspections combined with basic sensor networks to detect issues such as strain, displacement, and temperature changes. These methods often required physical access to the structure, lacked real-time insights, and were limited in their ability to predict future deterioration. While effective for identifying visible damage, traditional approaches were reactive rather than proactive, leaving critical gaps in long-term infrastructure management.

In contrast, when integrated with real-time SHM, DT systems offer a proactive solution. By coupling sensor-derived data (e.g., strain, displacement, vibration) with simulation models, DT platforms can evaluate structural behaviour under various scenarios, including load increases, material fatigue, and extreme environmental events. This capability allows infrastructure managers to forecast deterioration trends, simulate maintenance interventions, and prioritise repairs based on risk (Hakimi et al., 2023).

A key enabler of this predictive capability is the integration of high-fidelity 3D geometric models, generated via drone photogrammetry or LiDAR scanning, with live sensor feeds. These models provide a spatially accurate, dynamic representation of structural condition, which supports decision-making and facilitates remote assessment. In our framework, this approach was successfully applied to a heritage steel truss bridge, demonstrating how digital replicas can be used for both engineering evaluation and heritage preservation.

As SHM and DT technologies continue to evolve, future systems are expected to incorporate more sophisticated machine learning techniques for pattern recognition, anomaly detection, and risk forecasting. These innovations will improve system efficiency and enable scalable deployment across a broader range of legacy infrastructure. Ultimately, the integration of DTs into asset management strategies holds significant potential for extending service life, reducing maintenance costs, and preserving historically substantial structures sustainably and intelligently.

2.4. Applications of geospatial data in DT for aged infrastructure

Geospatial data plays a critical role in DT applications for infrastructure health monitoring, particularly when dealing with aged assets. By providing spatial context, geographic accuracy, and environmental metadata, geospatial data enhances the fidelity of structural models and supports more effective monitoring, maintenance planning, and decision-making (Mahmoodian et al., 2022).

When integrated with SHM systems, geospatial information enables the spatial localisation of sensors, accurate mapping of infrastructure conditions, and correlation with environmental variables. This is especially important for legacy assets, where location-specific deterioration patterns, such as corrosion due to humidity or fatigue near high-traffic zones, can be overlooked in non-spatial models.

Recent approaches have demonstrated the utility of geospatial tools in optimising SHM deployments. For example, Geospatial Network Analysers (GNA), developed using Python BIM libraries, support the analysis and optimisation of large-scale sensor networks by evaluating geographic dispersion and communication efficiency (Tariq et al., 2019). Integration of Building Information Models (BIM) with Geographic Information Systems (GIS), particularly using extended Industry Foundation Classes (IFC), enables real-time 3D visualisation, scenario simulation, and stakeholder engagement within immersive monitoring environments (Hakimi et al., 2023).

Geospatial data is also critical in coupling environmental monitoring with structural behaviour analysis. Techniques such as UAV-based LiDAR scanning and drone photogrammetry offer centimetre-level precision in geometric reconstruction, which is essential for deformation monitoring and change detection. Merging LiDAR with photogrammetry further improves the accuracy of Digital Elevation Models (DEMs), especially in complex terrains such as mountainous or post-glacial regions (Escobar Villanueva et al., 2019), (Chiang et al., 2017), (Šašak et al., 2019).

In our framework, 3D photogrammetry models of the heritage railway bridge were geospatially referenced and visualised via the CesiumJS engine using the 3DTiles format. This allowed stakeholders to interact with the structure in real-time, overlay sensor data, and simulate potential failure scenarios. This level of integration bridges the gap between spatial data management and structural analytics, offering an intuitive, high-fidelity platform for condition monitoring.

As DT systems evolve, geospatial data will remain central to their development. Its ability to contextualise structural performance in space and time makes it a powerful enabler of predictive maintenance, lifecycle optimisation, and asset-level risk assessment. By integrating geospatial analytics with SHM and simulation models, DTs can provide a comprehensive, location-aware foundation for managing ageing infrastructure in a dynamic and sustainable manner.

2.5. Challenges facing SHM-DT integration

Integrating SHM systems with DT platforms presents several technical, operational, and organisational challenges, particularly when applied to ageing and heterogeneous infrastructure systems. Despite advancements in various industries, DT systems are still in their nascent stages for large-scale infrastructure management. One of the primary obstacles is the complexity of data integration. This seamlessly combines diverse data types, including geometric models, real-time sensor streams, historical records, and environmental data, into a unified and functional platform. Ensuring compatibility, standardisation, and accuracy across these disparate data sources is crucial for reliable DT deployment (Chen et al., 2024).

A second challenge relates to the immaturity of DT technology for civil infrastructure. Unlike sectors such as aerospace or manufacturing, there is no universally accepted methodology for embedding SHM data within DT frameworks. Current implementations are often bespoke, and sensor limitations, data quality, and modelling complexity constrain the fidelity of predictions. This makes large-scale adoption difficult and highlights the need for standardised approaches.

Data management at scale presents another barrier. Continuous monitoring of large assets generates high-frequency data streams that require efficient storage, processing, and interpretation. While signal processing techniques such as Kalman filtering and wavelet transforms can reduce noise and enhance data quality, existing computational tools often struggle with the volume and velocity of data, hindering real-time decision-making. (Das et al., 2016).

Furthermore, the integration of physics-based and data-driven

models presents its own set of complexities. While FEMs are critical for simulating structural behaviour, they often rely on assumptions that may not reflect real-world conditions over time. Conversely, while capable of real-time anomaly detection, data-driven ML models require extensive, high-quality data for training. The practical and robust combination of these approaches into a hybrid model, leveraging their complementary strengths, remains an active area of research (Ye et al., 2019b).

Finally, a critical issue is the lack of comprehensive validation frameworks. As DT technology for SHM is not yet fully mature, there is limited consensus on how to rigorously validate its accuracy and performance. Continuous calibration of models using real-time data is essential, yet errors in this process can lead to inaccurate assessments of a structure's condition. Without established validation and verification processes, the reliability of DT systems for mission-critical applications, where safety and risk are paramount, remains a concern.

In summary, SHM–DT integration is challenged by data interoperability, technological immaturity, data management, hybrid modelling complexity, and insufficient validation methods. Addressing these barriers is essential to move from experimental deployments toward robust, scalable systems capable of transforming infrastructure management and eventually enhancing the resilience of infrastructures.

3. Hybrid system calibration outline

The calibration and validation of the proposed SHM-DT framework are essential to ensure that the digital model remains aligned with the physical structure and maintains high predictive accuracy and reliability. The process consists of two main stages: (1) Initial calibration phase, which aligns the FEM and 3D photogrammetry model to the bridge's as-is state, correcting for decades of wear and environmental effects, and (2) Continuous real-time update phase, which incorporates live sensor data to update the model parameters in real time, enabling early detection of degradation.

FEM Parameter Estimation: FEM parameters are updated through an Extended Kalman Filter (EKF), which recursively predicts system states and corrects them using accelerometer and strain measurements. This process reduces sensor noise, stabilises modal frequencies, and ensures the FEM reflects observed behaviour (Liu et al., 2023a), (Brenner et al., 2023). This process can be expressed mathematically as:

$$X_k = X_k^- + K_k\left(Z_k - HX_k^-\right) \quad (1)$$

where X_k is the updated state vector, X_k^- is the predicted state, Z_k is the measurement vector, H is the measurement matrix, and K_k is the Kalman gain, optimised to minimise the state estimation error covariance.

AI Module: Residual Learning and Adjustment: Future extensions include AI-driven residual learning to complement physics-based models. For example, CNNs may detect defects in imagery, while LSTMs can analyse vibration data for anomalies, enhancing sensitivity to non-linear behaviour (Bao and Li, 2020). It should be noted that the current demonstrator is limited to FEM–sensor–photogrammetry integration. The AI/ML modules (e.g., CNNs, LSTMs, and residual learning) are not yet implemented and are presented here as part of the future development roadmap.

Model Synchronisation and Feedback: Deviations between FEM predictions and measured data trigger parameter updates or alerts. EKF-based updating and predictive analytics allow the model to adapt to changes such as stiffness loss, cracks, or deflections while forecasting future performance (Torzoni et al., 2024).

Visualisation and Scalability: Updates are presented via a BIM-based 3D interface with sensor overlays. Cloud and edge computing ensure scalability and responsiveness, enabling real-time decision support and long-term asset management (Hao et al., 2024).

The hybrid calibration mechanism described above relies on accurate and timely data streams from the physical structure. The following section describes the underlying SHM system architecture responsible for delivering these streams through a multi-layered pipeline. In a simple flowchart, Fig. 2 illustrates the proposed real-time feedback loop, showing the continuous cycle of data processing, modal identification, FEM comparison, calibration, threshold and model updates.

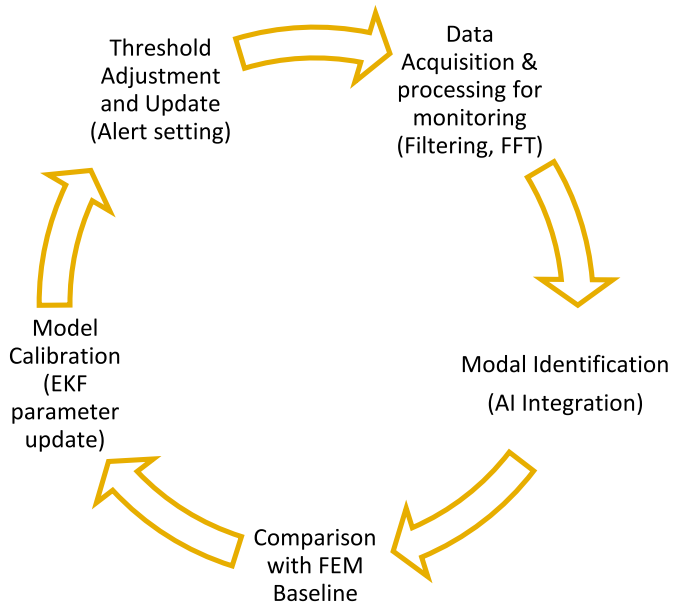


Fig. 2. Real-time feedback loop for DT-SHM framework.

4. Digital twin framework: layered system design for SHM integration

The SHM system architecture serves as the operational foundation of the proposed DT platform, enabling real-time sensing, data analytics, and feedback integration. It comprises five layers, from raw sensor data collection to decision support, designed to facilitate reliable, modular, and scalable structural behaviour monitoring. While the DT framework performs continuous calibration through AI and FEM coupling, the SHM architecture ensures uninterrupted data flow and analytic readiness across spatial and temporal domains.

The multi-layered design integrates advanced technologies such as geospatial data, photo-mesh 3D modelling, Finite Element analysis, and IoT-based real-time monitoring. Together, these layers support predictive maintenance, optimise structural management, and provide a pathway for broader DT adoption in infrastructure resilience management. Fig. 3 illustrates the hierarchical layers of the DT, presenting a simplified visualisation of its multi-layered architecture. The following subsections detail each layer of the proposed DT framework, outlining its purpose, data sources, tools, and integration strategies.

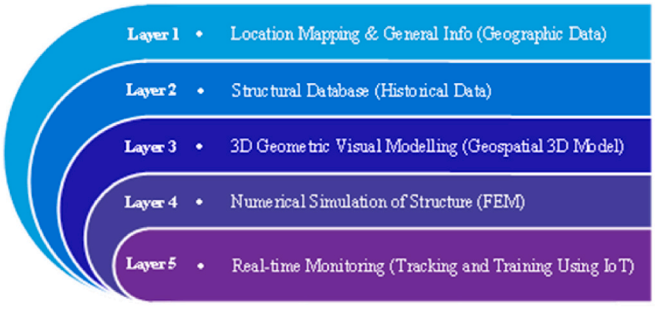


Fig. 3. DT Demonstrator's different layers.

4.1. Layer 1: location mapping and general information

The first layer of the DT system focuses on geographic mapping and metadata aggregation. It involves compiling location-based data for road and rail bridge infrastructures across Australia, integrated into an interactive geospatial platform. Key details about the structures, including bridge type (e.g., bridge/culvert/pedestrian), usage (i.e., rail/road), material (i.e., concrete/steel/timber) construction year, and ownership (i.e., councils, state government, Federal government), are presented in an accessible and categorised format, offering a comprehensive geographic overview. This layer provides essential contextual information for decision-making and strategic planning. The data for this layer is sourced from open-access government and state-level databases, ensuring consistency and broad coverage across various authorities. The bridge data are collected from the government data set for Victoria (VIC), New South Wales (NSW), Australian Capital Territory (ACT), Western Australia (WA), South Australia (SA), Queensland (QLD), and Tasmania (TAS). While this layer does not incorporate advanced technological tools, it is a critical baseline for more detailed structural monitoring and analysis in the subsequent layers. Fig. 4 illustrates the constitution of the foundational layer (Layer 1) of CSDILA's DT Platform, which localises Australian national transport infrastructures for indicative bridges. The bridges are colour-coded and categorised into different classifications, such as functionality (road bridge, rail bridge, pedestrian bridges, and viaduct, etc), structure type (beam, arch, truss, suspension, cantilever) bridges, and material properties (concrete, steel, timber, masonry, etc), ensuring detailed and systematic organisation for practical analysis and application.

4.2. Layer 2: historical and structural data collection

Layer 2 of the DT platform functions as a consolidated database, storing historical and structural information for each bridge, which is essential for lifecycle management and informed decision-making. This layer compiles data such as archival imagery, engineering drawings, inspection reports, maintenance records, material properties and design load specifications, facilitating a comprehensive understanding of a bridge's structural history, operational context, and condition evolution. Such records are crucial for bridges' health monitoring, guiding conservation strategies, and compliance with historical standards. Inspection reports and deficiency logs provide detailed assessments of structural issues such as cracks, corrosion, deformation, or joint wear, tracking deterioration trends over time and enabling proactive maintenance strategies.

This layer also incorporates maintenance records and retrofitting histories, thus providing a sequential narrative of past interventions and modifications. These data are instrumental in understanding how repairs or retrofits have altered the structural behaviour of the bridge, influencing its current and future performance under operational loads. In addition, this layer supports dynamic updates, allowing recent inspection data, photographs, and reports to be added to the repository. Updated data helps to identify changes in structural features, such as new cracks, discolouration, or deformation, ensuring that the platform reflects the most current condition of the bridge.

This layer incorporates operational and environmental data, recording prolonged exposure to traffic patterns, climate conditions, and extreme events such as floods or seismic activity. Including this data is crucial for assessing the cumulative damage and forecasting the structure's future responses. A standardised metadata framework is used to integrate with other layers of the DT platform, organising a diverse dataset for efficient analysis and accessibility. Fig. 5 illustrates the architecture of Layer 2 in the DT Platform, showcasing its role as a comprehensive repository for historical, structural, and operational data.

4.3. Layer 3: 3D visual modelling and geometric assessment (data layer)

This layer utilises advanced drone-based photogrammetry and laser scanning technologies to generate detailed three-dimensional photo mesh models and point cloud datasets, providing a comprehensive geometric representation of bridge infrastructure. Photogrammetry and 3D photo mesh modelling serve as the visual and spatial foundation of the DT, providing highly accurate geometric representations of the structure (Wang et al., 2022). These models facilitate the visualisation of structural components, offering stakeholders an effective tool for conducting remote condition assessments and geometric evaluations. The photo mesh models deliver detailed visualisations, whereas the point cloud datasets, comprising millions of spatially referenced data points, capture intricate geometric details and dimensions, ensuring a robust geometric foundation for advanced structural analysis.

Point cloud data, obtained through technologies such as light detection and ranging (LiDAR) and advanced photogrammetry software, ensures that even precise features of the bridge are captured accurately. This data is essential for the detection of geometric anomalies, deformation patterns, and structural deterioration over time. The integration of these technologies allows for the development of spatially accurate and up-to-date models, which serve not only as a reference for monitoring but also as the basis for further analytical modelling, including finite element analysis and sensor placement strategies. This layer thus bridges the gap between visual representation and analytical precision, thereby supporting comprehensive infrastructure management and planning.

UAVs, commonly called drones, play a crucial role in capturing high-resolution images of bridges from multiple angles. These images undergo advanced photogrammetric processing using software such as Agisoft and Metashape, which generate precise 3D point cloud models.

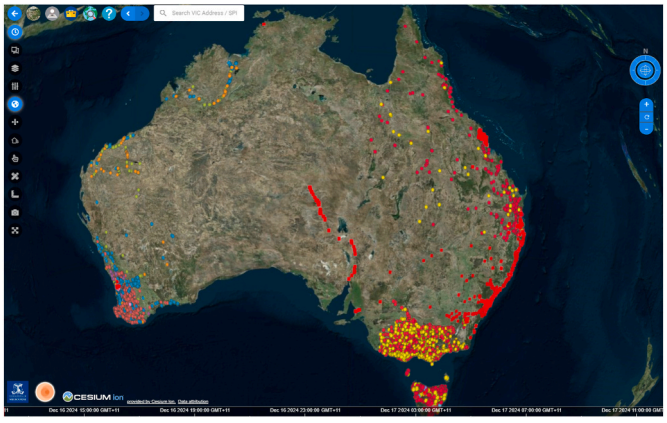


Fig. 4. DT platform; Layer one (Bridge Localisation).

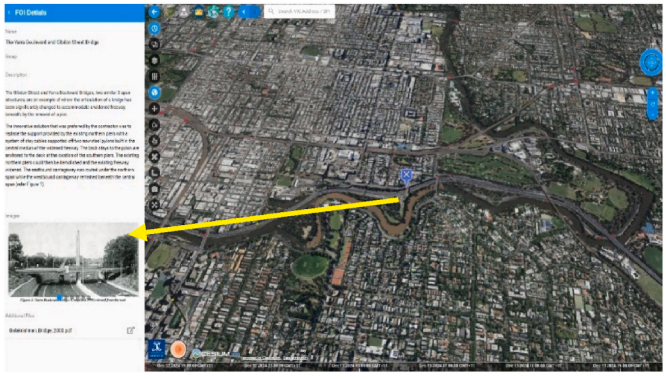


Fig. 5. Architecture of Layer 2 in the CSDILA's DT platform, layer 2 consolidating historical and technical information.

The point cloud data are then refined into a detailed mesh, with textures applied to create an accurate, high-fidelity digital representation of the bridge. This meticulous process ensures that critical structural elements, including joints, trusses, boundary conditions and intricate architectural details, are accurately preserved, making them particularly valuable for historical structures and complex geometries.

Beyond serving as a static visual representation, these 3D models form the geometric foundation for further structural analysis and DT integration. The drone-based approach enables comprehensive coverage, including hard-to-reach areas of the bridge that are traditionally difficult to inspect manually. This level of precision and accessibility is particularly beneficial for long-term monitoring, ensuring that even minor structural changes, such as deformations or material degradation, are captured over time.

To enhance visualisation and interoperability within the DT platform, the 3D mesh models are converted into the 3DTiles format, a spatial data format optimised for efficient streaming and rendering of large-scale geospatial datasets. This conversion process segments the high-resolution 3D models into hierarchical levels of detail, allowing dynamic loading based on the user's viewpoint and computational capacity. The 3DTiles dataset was rendered using CesiumJS, a powerful web-based visualisation engine that enables real-time interaction with geospatially accurate digital models. Through CesiumJS, users can seamlessly navigate, inspect, and analyse bridge structures in an immersive 3D environment, facilitating remote condition assessments and enabling decision-makers to visualise structural integrity at various scales. This integration of photogrammetry, 3DTiles, and web-based rendering significantly enhances the accessibility and usability of bridge monitoring data. By embedding the models into an interactive geospatial platform, stakeholders, including engineers, asset managers, and policymakers, can remotely assess structural conditions, detect anomalies, and make informed decisions without requiring on-site visits. The combination of high-resolution modelling and interactive visualisation establishes a robust foundation for integrating real-time sensor data and numerical simulations, ensuring a dynamic and continuously updated DT for structural health monitoring. Fig. 6 illustrates the integration of 3D photo mesh models within the DT platform, demonstrating how these visual representations serve as the foundation for further structural and analytical modelling.

4.4. Layer 4: ***numerical simulation using FEM*** *(visualisation)*

Layer 4 provides the analytical foundation of the DT framework, leverageing FEM to simulate structural behaviour under various operational and environmental conditions. The FEM predicts modal frequencies, deflections, and stress distributions, offering insight into load paths and structural response. However, FEM accuracy depends on assumptions about material properties and boundary conditions, which may become less valid as structures age (Fig. 7).

To maintain reliability, FEM parameters such as stiffness and damping are updated through data assimilation techniques, including the EKF and Bayesian updating. These methods recursively refine model parameters using sensor data, reducing prediction error and aligning simulations with observed responses. The mathematical formulation of the EKF involves two main steps:



Fig. 6. Layer 3 of the DT platform, integrating a 3D photo mesh model.

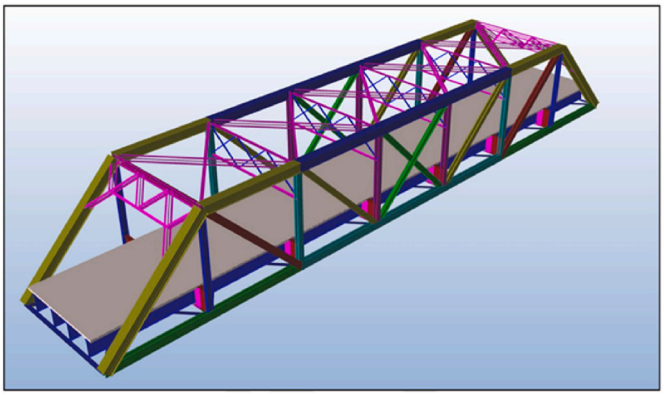


Fig. 7. FE model of the middle span of the bridge created using SpaceGass v14.2.

1. Prediction Step:

$$\hat{x}_{k|k-1} = f\left(\hat{x}_{k-1|k-1}\right) \tag{2}$$

$$P_{k|k-1} = F_{k-1} P_{k-1|k-1} F_{k-1}^{T} + Q_{k-1} \tag{3}$$

2. Update Step:

$$K_k = P_{k|k}\, H_k^{T}\left(H_k H P_{k|k-1} H_k^{T} + R_k\right)^{-1} \tag{4}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\left(y_k - h\left(\hat{x}_{k|k-1}\right)\right) \tag{5}$$

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \tag{6}$$

where x is the state vector of FEM parameters, P is the error covariance matrix, H is the measurement matrix, and y is the observed data. This iterative process creates a feedback loop that continuously refines the FEM, enhancing its predictive capabilities and enabling proactive maintenance strategies.

In vibration-based systems, Reduced-Order Modelling (ROM) projects the high-dimensional FEM into a lower-dimensional subspace, enabling near-real-time updates without compromising reliability (Ozbulut et al., 2015). The mathematical representation of this projection can be expressed as:

$$\{u\} \approx [\Phi]\{q\} \tag{7}$$

where {u} contains the displacement vectors, [Φ] is the matrix of mode shapes and {q} is the reduced coordinate vector. This approach significantly reduces computational demands, enabling the framework to handle large-scale structures effectively.

The integration of FEM into the DT establishes a continuous feedback loop that enhances monitoring and predictive capabilities. Real-time updates improve accuracy, enable early detection of issues, and support targeted maintenance strategies. By fusing physics-based simulations with data-driven updating, the hybrid framework provides a robust platform for real-time calibration and predictive maintenance (Fig. 8). Future work will focus on incorporating advanced ML techniques to further improve predictive accuracy and scalability.



Fig. 8. Layer 4, Integration of FE model into the DT platform.

4.5. Layer 5: interoperability and integration

Layer 5 transforms the DT platform into a dynamic monitoring system by integrating data streams from IoT sensors deployed across bridge infrastructure. By incorporating real-time data into a digital model or FEM, this layer enhances predictive capabilities and enables proactive maintenance strategies.

The sensor network includes devices like accelerometers, strain gauges, displacement monitors, and temperature sensors. These data streams provide a holistic, real-time picture of the asset's structural health, allowing for informed engineering decisions. Data are transmitted wirelessly to a centralised processing system via communication protocols such as LoRaWAN or cellular networks. LoRaWAN operating on low-power unlicensed frequency bands (e.g., 868 MHz in Europe, 915 MHz in Australia and North America) supports energy-efficient, long-range communication, which is ideal for infrastructure located in remote or hard-to-reach areas.

By comparing live sensor data with defined baseline simulations from Layer 4, the system detects deviations such as abnormal stress patterns, excessive displacements, or changes in modal frequencies. These anomalies can trigger automatic alerts, enable timely maintenance interventions and prevent critical failures. As a result, Layer 5 strengthens the DT framework's ability to evolve alongside the physical asset, maintaining an accurate and continuously updated digital reflection of the structure under both operational and environmental conditions.

The detailed internal mechanisms that enable this real-time structural intelligence, spanning data acquisition, modelling, and feedback, are described in the following subsection on the SHM architecture.

4.5.1. SHM architecture

The SHM architecture serves as the dynamic engine of Layer 5 within

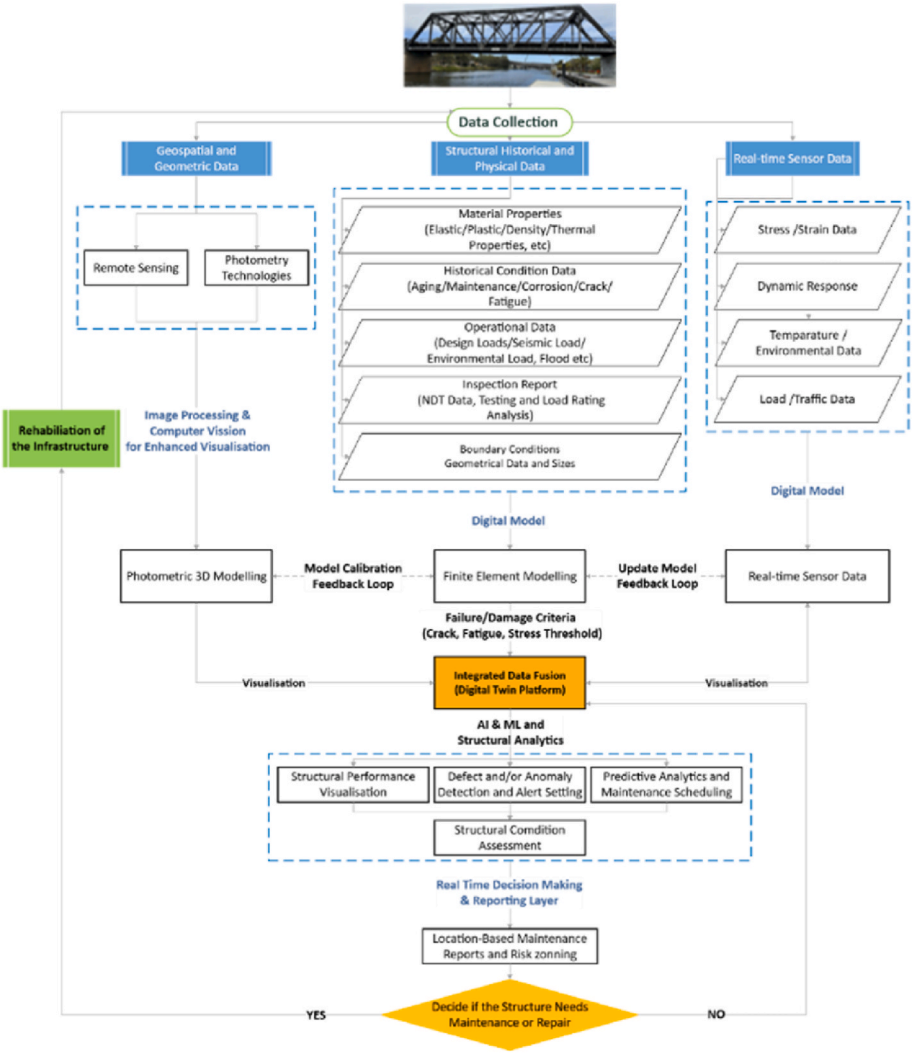


Fig. 9. Structural Health Monitoring architecture within the Digital Twin framework.

the proposed DT framework. It interprets raw sensor inputs into actionable insights through a closed-loop system that drives near real-time model updates and predictive analytics. Designed as a modular, scalable and interoperable system, the SHM architecture integrates data acquisition, modelling, numerical modelling, and feedback control to maintain synchronisation between the physical structure and its digital representation.

Whereas SHM specialises in sensor-level data collection and immediate anomaly detection, the DT elevates these outputs by integrating them with historical data, FEM simulations, and AI-powered analytics. Parameters such as stress, strain, vibration, displacement, and temperature are continuously monitored, providing granular and dynamic information crucial for maintenance prioritisation (Bao and Li, 2020). Together, SHM and DT form a synergistic system that bridges operational responsiveness with long-term strategic planning, ensuring asset safety, longevity, and resilience.

Fig. 9 illustrates the core components of the SHM system, structured in four interconnected stages:

1 Data Acquisition:

The process begins with data capture through advanced sensing technologies. This includes:

- ○ Geospatial data: UAV-based photogrammetry and laser scanning for geometric modelling
- o IoT sensors: Accelerometers, strain gauges, temperature sensors, and a displacement monitor for real-time dynamic input

These technologies enable a hybrid acquisition of both static (geometry and material properties) and dynamic (operational) parameters, ensuring comprehensive coverage of the structure's current state.

2 Integration with Numerical Modelling:

The collected data is mapped to a physics-based FEM for simulation. The 3D photogrammetry models provide geometric fidelity, while FEM simulates responses to operational loads, environmental conditions, and structural anomalies. This step establishes the computational backbone of the DT and forms the basis for iterative calibration.

3 Numerical Feedback Loop:

To ensure synchronisation between the physical asset and its digital representation, a real-time feedback mechanism is established. Assimilation algorithms, such as the EKF, RLS, or Bayesian Inference, continuously adjust FEM parameters (e.g., stiffness, damping, boundary conditions) using incoming sensor data. Modal analysis plays a key role, with extracted frequencies and damping ratios serving as indicators of stiffness loss or damage accumulation.

4 Decision Support and Predictive Analytics:

The calibrated DT outputs are visualised through a BIM-integrated interface, allowing stakeholders to track structural condition, assess future scenarios, and plan maintenance. Advanced AI techniques, such as CNNs for crack detection and LSTMs for anomaly forecasting, complement this system by recognising hidden patterns and predicting emerging structural risks.

This closed-loop SHM-DT system supports real-time condition monitoring, early anomaly detection, and long-term infrastructure resilience. While SHM focuses on detailed sensor-level analysis and operational insights, the DT elevates these outputs into a strategic layer for planning, simulation, and stakeholder engagement. The effectiveness of this SHM–DT integration has been demonstrated through its deployment on a historically significant railway bridge. The following section presents this case study, outlining sensor deployment, model calibration, and preliminary findings from real-time structural monitoring.

5. Demonstration and field implementation of DT of a railway bridge for SHM

This section demonstrates the implementation of the proposed DT framework on a steel railway truss bridge located in Australia. The framework integrates an FEM, drone-based 3D photogrammetry, and real-time IoT sensor data transmitted via LoRaWAN. The aim is to establish a closed-loop system enabling continuous structural health assessment, anomaly detection, and model updating.

5.1. Bridge context and FEM modelling

The case study was conducted on a heritage steel truss railway bridge in Australia, constructed in the late 1920s. The bridge consists of two plate girder approaches and a 50 m simply supported truss span, which remains a critical freight corridor today. Its historical significance and high operational demand make it an ideal candidate for DT–SHM.

A representative FEM was developed using *Space-Gass v14.2*, with material and boundary conditions calibrated against 1927–1929 design specifications and ARTC inspection data (S. Australia, 2017). The model prioritised computational efficiency while representing axial, shear, and bending responses. A UAV-based 3D photogrammetry model was generated and aligned with the FEM, ensuring geometric fidelity and providing a realistic visualisation environment within the DT platform (refer to Fig. 10).

5.2. Wireless sensor layout and configuration

To capture the bridge's dynamic behaviour, eight triaxial wireless accelerometer nodes were installed at strategic locations on the truss (e.g. mid-span chords and critical joints). Sensor placement was guided by an FEM modal analysis of the bridge, targeting points of high mode shape amplitude to maximise information gain (Armijo and Zamora-Sánchez, 2024). Each sensor is a LoRaWAN-enabled MEMS accelerometer (Move Solutions Gen-3), with a ±2g range and high sensitivity suitable for ambient vibration monitoring (Armijo and Zamora-Sánchez, 2024). The nodes are time-synchronised and assigned unique Device EUI identifiers for the LoRaWAN network, ensuring each data stream is correctly associated with its sensor node. Sensor placement is illustrated in Fig. 11.

Each accelerometer node is configured in a dual-mode acquisition scheme: a low-power programmed sampling mode and an event trigger-based or threshold sampling mode. In programmed mode, the sensors wake up at fixed intervals (varying from 1 to 6 h; in this scenario, it is set for 1 h) to record ambient vibrations. Event-triggered mode is used whenever a significant vibration event is detected. The trigger is governed by acceleration threshold settings described below. This hybrid scheme balances data quality and battery life, capturing important transient events (like train passages) while still tracking long-term



Fig. 10. 3D photo mesh model of the bridge created using drone photos, including side view and top view.

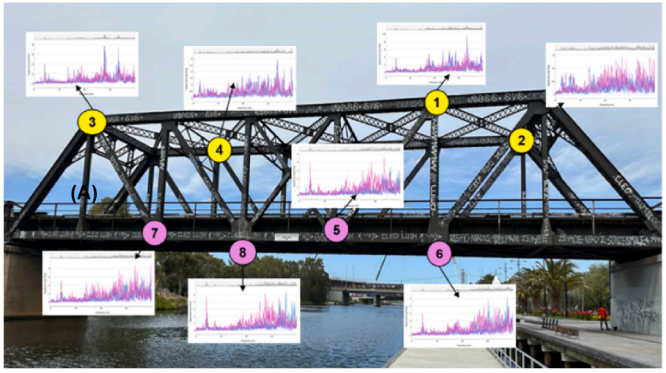


Fig. 11. DT visualisation with accelerometer locations and (A) Sample frequency response of Sensor 7.

behaviour. Sensor synchronisation across the span is accurate to within milliseconds, allowing simultaneous multi-point response measurement.

These correspond to moderate and extreme vibration events, respectively. If the peak acceleration during an event exceeds 100 milli-g, a warning flag is raised in the DT dashboard, suggesting the need for inspection; if 500 mg is exceeded, an automatic critical alert is issued to stakeholders. The threshold approach leverages the LoRaWAN sensor's design, and when an event threshold is exceeded, the sensor immediately transfers the data to the gateway. This ensures that significant vibration events (like fast heavy trains or potential impacts) are not missed, while routine low-level vibrations do not overload the communication system. All thresholds were determined considering both the sensor specifications and historical records of bridge acceleration.

5.3. Sensor sampling frequency & bridge vibration alert thresholds

To support dynamic monitoring of the heritage bridge, the wireless accelerometers were configured with a multi-level threshold system designed to classify vibration severity and trigger appropriate acquisition and alert responses. These thresholds classify vibration severity and trigger appropriate actions, from data recording to alerts. The values were selected based on initial system tests, sensor specifications, and historical vibration data, with an emphasis on the bridge's age and avoiding false alarms. As further data is collected and analysed, these thresholds will be reviewed and adjusted if necessary (the detailed recalibration analysis is ongoing).

The accelerometers sample at 80 Hz frequency (which is sufficient to capture the bridge's vibration frequencies well beyond the first few modes). All raw acceleration time-histories are time-stamped and transmitted after each event for analysis. A conservative activation of 30 mg (0.030 g) threshold was selected following field tests, providing a conservative trigger to capture even small dynamic events, particularly given the bridge's age and structurally sensitive joints.

Initially, the "Alert" (100 milli-g), "Warning" (200 milli-g), and "Critical" (500 milli-g) thresholds were set based on both sensor capabilities and expected bridge behaviour. At this stage, these values were set conservatively to account for structural uncertainties and to minimise the risk of missed detections or false negatives. Importantly, these threshold values are not static; they are considered provisional and will be refined through the feedback loop once sufficient operational data is collected and the DT is calibrated. This tiered threshold strategy, which starts with a low activation sensitivity (30 milli-g) for reliable event triggering, followed by escalating alert levels, achieves a balance between monitoring precision and communication efficiency. Fig. 12 illustrates the operational dashboard of the installed SHM system, confirming all sensors and gateways were online and actively reporting data during the monitoring period.

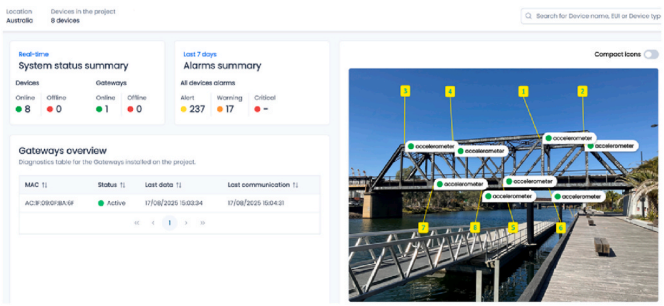


Fig. 12. Real-time system status dashboard for the installed SHM–DT system.

5.4. Digital twin platform and sensor integration

The DT platform, built upon CSDILA's platform, acts as a central hub where the physical data and numerical model, FEM, continuously interact and are calibrated. A photogrammetry-based 3D mesh serves as the visual foundation, onto which the Space-Gass FEM is overlaid using a shared spatial reference system for accurate alignment. LoRaWAN-enabled accelerometers are registered in the DT environment, and their unique identifiers (DevEUIs) link incoming data to precise physical locations on the virtual bridge. As sensor streams are received, key parameters, such as acceleration magnitudes, temperature, and battery status, are mapped onto the 3D interface for intuitive visualisation and spatial context.

Data transmission is handled via a secure LoRaWAN gateway connected to a cloud-based platform, which exposes a RESTful API for structured data access. The DT platform uses this API to retrieve sensor data in real time, ensuring a continuous flow of up-to-date information. While the current system focuses on real-time condition tracking and sensor status visualisation, deeper analytics and numerical calibration, such as modal correlation and FEM updating, are implemented in a separate pipeline described in the next section. This layered architecture allows the DT to serve as a live, interactive hub where spatial models and field measurements converge to support operational awareness and asset management.

5.5. Real-time feedback loop mechanism

A key feature of the DT–SHM framework is the real-time feedback loop between sensor measurements and the FEM model, ensuring the DT remains accurate and continuously self-updates with incoming data (Wang et al., 2025). After each significant vibration event (e.g. a train crossing), the system performs an FFT on the recorded acceleration data to identify the dominant modal frequencies of the bridge. The fundamental frequency and higher modes appear as peaks in the FFT amplitude spectrum. In initial tests, for example, the first three modes were found around 2.3 Hz, 4.6 Hz, and 7.9 Hz, aligning well with the baseline FEM's predicted vertical and torsional modes (Jeon et al., 2025). These identified frequencies are automatically logged for trend tracking.

The measured modal frequencies are then compared to the baseline FEM's modal analysis results. If a measured frequency deviates from the FEM-predicted value by more than a set tolerance (e.g. > 5 %), the system flags a discrepancy. Such shifts in natural frequency can indicate changes in structural stiffness or boundary conditions. For instance, a drop in frequency often suggests a stiffness reduction (potential damage or loosening), whereas an increase might indicate an unexpected boundary stiffening or simply a sensor calibration error. This vibration-based change detection follows well-known SHM principles, where changes in modal properties signal structural changes (Wang et al., 2025). The platform continuously monitors these differences in real time.

Whenever a significant discrepancy is detected, the DT triggers an automated model update. A straightforward approach in our project is to

uniformly adjust the bridge's stiffness so that the FEM's fundamental frequency matches the measured frequency. This leverages the relationship between frequency and stiffness ratio of the system ($f \propto \sqrt{K/M}$, meaning natural frequency rises or falls with the square root of stiffness). In practice, this can be implemented by scaling the global stiffness matrix by:

$$K_{new} = K_{old} \times \left(\frac{f_{measured}}{f_{FEM}}\right)^2 \quad 8$$

More advanced calibration methods are also common; for example, sensitivity-based optimisation or Bayesian model updating can adjust individual element parameters to minimise the error across multiple mode frequencies. In our current demonstration, the simpler frequency-matching approach was chosen for its robustness and speed, enabling near real-time updates. After the FEM properties are updated, a new modal analysis is run to produce updated mode shapes and internal force distributions, ensuring the digital model better reflects the physical structure's current state. All changes are logged, and if any adjusted parameter exceeds preset limits, the system issues an alert that potential damage may have occurred, prompting further engineering inspection.

Basically, the DT and the physical structure remain in sync through a closed-loop learning system. This feedback process continuously refines the model based on the updated data. For example, if a particular joint gradually loosens over time, the fundamental frequency will correspondingly decrease over time; the DT platform would detect this trend and incrementally adjust the model to quantify the stiffness loss. The updated FEM can also generate new predictions (e.g. deflections or stresses under load), which can be cross-checked against additional sensors or periodic surveys for validation, thereby fusing sensor data with physics for greater confidence. Such a dynamic model-updating strategy is common in state-of-the-art SHM research and is here realised in an operational setting.

5.6. Data processing and visualisation

Sensor data from each event is automatically retrieved from the DT platform's database and processed using a Python-based pipeline developed for this project. Selected samples representing the bridge's free vibration response are analysed to estimate key modal parameters, including natural frequencies and damping ratios. Fig. 13 provides the strategy for data collection and calculation of modal parameters of the structure, calculating the frequency and damping ratio, and the validation procedure. Steps after data collection include (i) noise removal and filtering, (ii) extraction of free-decay segments, (iii) frequency-domain analysis for modal identification, and (iv) calculation of structural parameters (stiffness K, mass M, damping C) and damping ratio ζ. These outputs are used for FEM calibration, anomaly screening, and long-term trend analysis within the DT–SHM framework.

Each acceleration time-history needs to be detrended and filtered, with a high-pass filter (~0.1 Hz) to remove very low-frequency drift and a band-pass filter to isolate the bridge's vibration range (Fiandaca et al., 2022). Basic statistics such as peak values, root mean square (RMS), and record duration will be computed, and any anomalous segments will be flagged. Frequency content will be extracted via Fast Fourier Transform (FFT) with a Hanning window, enabling identification of dominant modes and qualitative mode shape assessment.

Results are visualised in both frequency and time domains through the DT platform, providing interactive plots for quick inspection of structural responses to train passages. Processed parameters will be stored in the system's database for long-term trend analysis, supporting early detection of potential stiffness loss or other structural changes. Alarm thresholds highlight significant events, ensuring rapid identification of exceedances.

5.7. Vibration data

The SHM system has been continuously recording and analysing the bridge's vibration events since installation, with typical events occurring during freight train passages. Eight triaxial accelerometers capture responses along the longitudinal (X), transverse (Y), and vertical (Z) axes, providing the basis for quantifying the bridge's dynamic behaviour under operational loading. In this case study, data were collected in 1-h intervals, with each programmed acquisition lasting 102 s at a sampling frequency of 80 Hz.

Prior to analysis, the raw signals are processed through standard automated filtering and detrending procedures. Synchronisation across all eight sensors was performed to ensure temporal alignment of events, which is critical for modal interpretation and comparative assessment. Representative time histories demonstrate clear transient responses corresponding to events and loads, with the variations in amplitude and frequency content evident across the top and bottom chord sensors.

Statistical descriptors such as peak acceleration, root mean square (RMS), and standard deviation were extracted to characterise the response intensity. Results consistently showed higher amplitudes at the bottom chord sensors, reflecting their proximity to the primary load paths, while top chord sensors exhibited comparatively attenuated responses. These variations provide valuable insight into load distribution and structural participation. Natural frequencies are identified as peaks in the power spectral density, with corresponding mode shapes derived from singular vector analysis. Damping ratios are estimated using the half-power bandwidth method:

$$\zeta = \frac{\Delta f}{2 f_n} \quad 9$$

where f_n is the natural frequency and Δf is the half-power bandwidth (Chopra, 2007), (Zhang et al., 2011). Free vibration segments of the events are used for evaluating the damping characteristics of the bridge. Preliminary damping is ~2.5–3 % for the first observed mode and ~3–4 % for higher modes; a mode-specific damping study is planned to

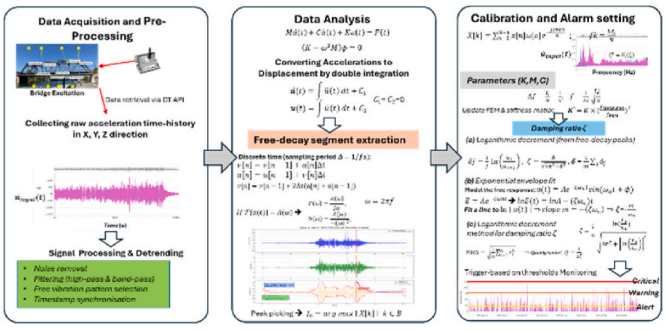


Fig. 13. Data processing workflow used to convert raw acceleration signals into modal parameters for frequency and damping estimation.

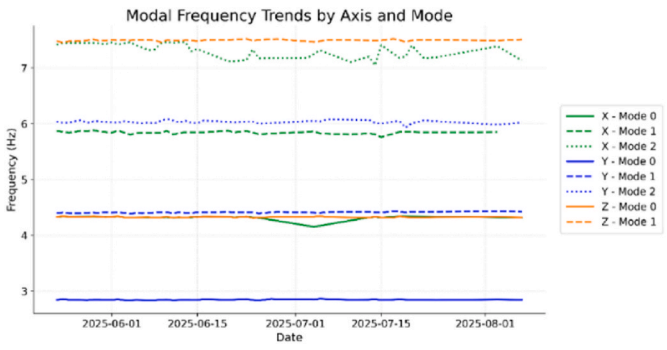


Fig. 14. Time history of the first three modal frequencies across X, Y, and Z axes from May–August 2025.

improve robustness. While the present analysis offers indicative results, a more detailed and mode-specific investigation of damping behaviour is planned for future studies to enhance the robustness of the monitoring framework.

Frequency-domain analyses further highlight dominant vibration modes, with the fundamental frequency consistently identified across multiple sensor locations. Fig. 14 presents the historical evolution and distributions of the first three modal frequencies (Modes 0–2) across X, Y, and Z axes from May–August 2025. The coefficients of variation (CV) are typically low (≤~0.5 %; see Table 2), indicating seasonal/operational stability and supporting FEM calibration. To better visualise the relative stability of modal frequencies across sensing axes, the CV values are presented as a heatmap in Fig. 15. The uniformly low CV values (<1 %) confirm modal stability across all modes. The slightly elevated variability observed for Mode 2 in the X-axis is consistent with the broader spread noted in the frequency distributions. Fig. 16 shows the corresponding probability density functions, with tight clustering around the modal means. To support model calibration, these frequencies will be investigated against finite element model predictions in a future study.

The baseline FEM predictions for the first modes were 1.964, 2.447, 2.473, and 5.492 Hz. In the measured spectra, only modes with enough excitation and participation at the instrumented DOFs were consistently identifiable; modes with weak operational excitation are less visible. Therefore, comparisons between FEM and measurements focus on clearly observed modes, and any discrepancies at lower frequencies are attributed to idealised boundary conditions and current sensor placement. These issues will be revisited as monitoring expands, and modal identification is further refined.

Across the study period, modal frequencies clustered around 2.8–4.3 Hz (Mode 0), 5.8–6.0 Hz (Mode 1), and ~7.0–7.5 Hz (Mode 2), compatible with the first truss deformation, a combined girder–truss mode, and a higher-order global mode, respectively. Minor variations (~0.05–0.10 Hz) are attributed to operational and environmental effects (train loading patterns, temperature). No persistent frequency drifts were observed over a six-month period, indicating no detectable stiffness loss. From aggregated free-decay segments, damping is estimated to be ~2.5–3 % for Mode 0 and 3–4 % for higher modes; these values will be incorporated into the updated FEM for calibration and sensitivity analysis.

5.8. Demonstration outcomes and system performance

The six-month field deployment provided quantitative evidence of the DT-SHM framework's operational viability under real-world conditions. The LoRaWAN sensor network demonstrated consistent operational performance throughout the monitoring period, with all eight accelerometer nodes maintaining continuous data transmission as confirmed by real-time dashboard monitoring of network connectivity and sensor status. The dual-mode acquisition scheme effectively balanced data capture requirements with power consumption

Table 2
Summary statistics for a selected sensor modal frequency recorded from May–August 2025, showing mean, standard deviation, and coefficient of variation (CV) for each mode and sensing axis.

Mode	Axis	Sample Count	Mean (Hz)	Std (Hz)	Min (Hz)	Max (Hz)	CV (%)
0	X	24.0	4.32	0.036	4.15	4.34	0.83
0	Y	47.0	2.85	0.007	5.93	6.09	0.26
0	Z	45.0	4.33	0.007	4.31	4.35	0.16
1	X	32.0	7.32	0.132	7.05	7.05	1.8
1	Y	43.0	4.41	0.010	4.39	4.43	0.23
1	Z	42.0	7.5	0.014	7.46	7.52	0.19
2	X	37.0	5.84	0.027	5.76	5.88	0.46
2	Y	42.0	6.03	0.028	5.93	6.09	0.46

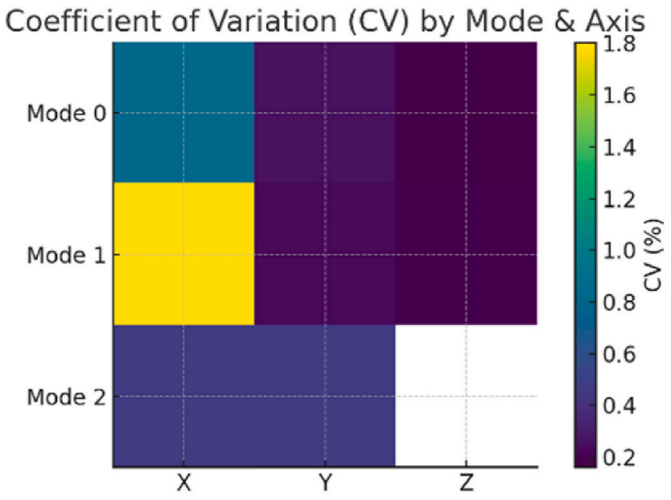


Fig. 15. Heatmap of CV (%) – coefficient of variation across modes and axes.

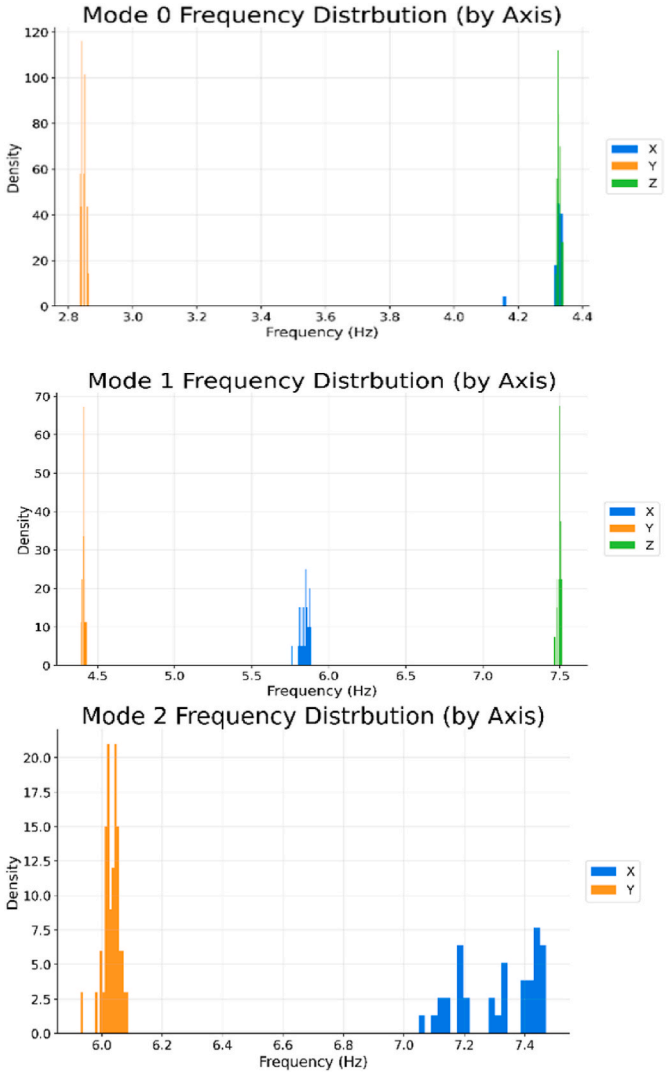


Fig. 16. Modal frequency distributions for Mode 0–2 derived from accelerometer data (May–Aug 2025); narrow spreads indicate stable modal behaviour.

constraints through programmed and event-triggered sampling protocols, supporting extended autonomous operation without manual intervention.

In the prototype phase, the tiered vibration thresholds (30 mg activation, 100 mg alert, 200 mg warning, and 500 mg critical) were initially defined conservatively based on sensor specifications and historical vibration records. During the six-month monitoring period, all major freight train passages were successfully captured, with no false critical alarms, confirming the operational validity of the chosen levels. These thresholds therefore provide a reliable baseline for deployment, though they remain provisional and will be further refined in future studies as additional long-term data and calibration results become available.

Statistical analysis of the extracted modal parameters showed strong time-based stability of the structure. Across the monitoring period, modal frequencies displayed coefficients of variation below ~0.5 %, indicating minimal drift due to seasonal or operational factors and providing a robust basis for model calibration.

To illustrate the calibration procedure, a preliminary comparison was made between FEM baseline predictions and measured modal frequencies. Analysing the FE model, the first three modes at 1.96, 2.45, and 2.47 Hz are calculated, whereas the observed operational responses (2.85, 4.4, 6.03 Hz) are higher. These discrepancies are likely related to idealised boundary conditions (e.g., pin and roller support) and material assumptions that differ from reality. Accordingly, the results are presented only as a proof of concept, demonstrating the feedback-loop workflow's capacity to align FEM predictions with field measurements and adjust the assumptions toward more realistic conditions. A full quantitative validation will be pursued in future works through extended monitoring, refined sensor configurations, and advanced modal identification techniques (e.g., FDD, SSI-COV).

The deployment confirms that the DT–SHM system can operate continuously on a live and operational bridge, delivering stable measurements, real-time visualisation, and preliminary model calibration. These results validate the framework's feasibility while highlighting the need for extended campaigns and more advanced analytics to achieve full quantitative validation.

5.9. Current limitations and future enhancement opportunities

While the current implementation successfully demonstrates the core DT-SHM concept, several methodological and operational limitations constrain its immediate functionality to broader infrastructure monitoring contexts. The present modal identification approach relies on elementary FFT peak-picking algorithms, which, although adequate for demonstration purposes, lack the sophistication of established operational modal analysis techniques such as Frequency Domain Decomposition (FDD) and Stochastic Subspace Identification (SSI-COV). Incorporating such methods could enhance automation and improve sensitivity to subtle structural changes that are not readily captured by fundamental frequency analysis.

Furthermore, while the system acknowledges environmental influences on structural response, comprehensive correlation analysis between ambient conditions (e.g., temperature, humidity), and measured modal parameters remains incomplete. This analytical gap limits the framework's capacity to distinguish between benign environmental variations and genuine structural degradation indicators.

The historical scope of the current validation presents additional constraints, as the six-month monitoring period, while sufficient for establishing operational feasibility, provides limited insight into seasonal behavioural patterns, long-term degradation trends, or the system's performance under extreme environmental conditions. Extended monitoring campaigns spanning multiple annual cycles would be necessary to fully validate the framework's reliability for persistent structural health assessment.

Systematic deployment across multiple infrastructure assets with varying characteristics would be essential to establish the framework's scalability and comparative performance across different bridge types and operational environments. It should be noted that the advanced analytical techniques referenced above remain in the conceptual and development stages, with all proposed enhancements requiring rigorous offline validation against historical datasets before integration into operational systems to minimise the risk of false positive alerts and ensure system reliability.

Finally, the practical deployment of a DT–SHM framework raises essential challenges related to cost, safety, and sensor configuration. Dense sensor networks, UAV-based surveys, and continuous data pipelines can substantially increase implementation and maintenance expenses, which may be prohibitive for agencies operating with limited budgets. In addition, the choice of sensors, along with decisions on their number, placement, and installation protocols, is highly consequential, as these factors directly influence data quality, monitoring coverage, and overall system reliability. Installation and monitoring activities must also comply with strict safety protocols, particularly in operational bridge environments where access can disrupt traffic and expose workers to potential hazards.

Successful deployment further depends on effective coordination among diverse stakeholders, including asset owners, regulators, maintenance contractors, and, in some cases, heritage authorities, to ensure that technical, legal, and cultural requirements are addressed. Robustness may also be compromised in constrained environments, such as remote areas with intermittent connectivity or limited power availability. Addressing these challenges will require not only cost-effective sensor strategies, optimised data collection methods, and energy-efficient communication networks, but also careful planning and stakeholder alignment to ensure that the framework is both technically viable and practically implementable.

5.10. Demonstration summary

The field implementation on the heritage railway bridge demonstrates that the proposed DT-SHM framework can be practically deployed and operated on complex, in-service infrastructure. The stable measurement performance and successful real-time data processing validate the technical feasibility of continuous structural health monitoring using LoRaWAN-enabled sensor networks. However, the demonstration scope remains limited by its focus on establishing operational capability rather than comprehensive validation of structural health assessment accuracy.

This work advances current practice by demonstrating end-to-end integration of diverse technologies within a unified digital twin framework. The successful deployment on active infrastructure provides empirical evidence that sophisticated monitoring systems can operate reliably without disrupting critical operations. The framework's modular architecture and cloud-based implementation indicate potential for scalable deployment across infrastructure networks, while the demonstrated measurement precision supports confidence in the approach's operational viability.

The demonstration establishes fundamental capability for continuous structural health monitoring rather than developing novel damage detection algorithms, representing a significant advancement toward intelligent infrastructure management. The stable modal frequency measurements and reliable system performance over six months provide confidence in the framework's potential for broader application. However, significant research opportunities remain to enhance analytical sophistication, expand temporal validation, and demonstrate broader applicability across diverse infrastructure contexts. The collected dataset provides a foundation for developing advanced modal analysis algorithms, environmental correlation models, and machine learning approaches for pattern recognition in structural response data.

6. Conclusion

The growing challenge posed by ageing infrastructure and the increasing complexity of asset management necessitates the development of advanced solutions for efficient monitoring, maintenance and management of bridge infrastructures. This study presents a pioneering implementation of a multi-layered DT framework for SHM applied to a heritage steel truss railway bridge in Australia. Through the integration of numerical modelling, drone-based photogrammetry, and real-time IoT sensor data transmitted via sensor networks, the framework establishes a robust foundation for continuous structural health assessment.

The methodology employed in this research involved a comprehensive analysis of modal parameters derived from ambient vibrations, utilising modal analysis techniques. These parameters, including natural frequencies and mode shapes, were extracted from real-time sensor data and compared against baseline FEM simulations. This iterative process enabled quantification of structural response variations and demonstrated the feasibility of automated model calibration to maintain synchronisation between the digital twin and physical structure.

The six-month field deployment successfully validated the framework's operational capability, demonstrating stable measurement performance with modal frequency coefficients of variation below 0.5 %. The implementation achieved consistent data transmission through sensor networks and successful real-time processing of structural response data. However, the demonstration scope remained focused on establishing operational feasibility rather than comprehensive validation of structural health assessment accuracy. Key innovations include the closed-loop feedback mechanism that integrates sensor data with physics-based modelling, enabling continuous model updating and providing a foundation for proactive maintenance strategies.

The findings underscore the transformative potential of the SHM-DT integration in the advancement and digitisation of infrastructure management. By bridging the gap between conventional assessment methods and next-generation predictive analytics, this framework offers a scalable and adaptable solution applicable to a wide range of infrastructure types, by focusing on more data interpretation for reliable decision-making. The successful application to an Australian historically significant railway bridge not only validates the framework's practicality but also highlights its relevance in preserving cultural heritage while ensuring structural integrity monitoring.

In summary, this study lays the groundwork for future advancements in SHM through the implementation of DT technologies, providing a vital tool for the proactive management of infrastructure. The insights gained from this research contribute to the ongoing evolution of infrastructure monitoring practices, promoting resilience and sustainability in the face of increasing operational demands and environmental challenges. However, transition to operational implementation requires addressing identified limitations through systematic research and expanded validation across diverse infrastructure contexts.

7. Future directions and opportunities

The proposed SHM-DT framework for bridges lays a strong foundation for future advancements in infrastructure management; however, its potential extends far beyond its current scope. As technology continues to evolve, several exciting opportunities have emerged to enhance the framework's capabilities, scalability, and applicability.

A key priority is the enhancement of analytical algorithms for the analysis of raw data and the interpretation of it into understandable data. Future work will embed the development of a solution algorithm, backup modelling, and advanced optimisation techniques directly into the DT pipeline, enabling faster model convergence, reduced computational load, and improved interpretability.

Another important avenue for future work will also be the refinement of the provisional vibration thresholds (30 mg, 100 mg, 200 mg, 500 mg). Extended monitoring and data-driven optimisation, potentially supported by ML methods, will allow thresholds to adapt dynamically to different structures and operating environments.

Looking ahead, the integration of ML & AI will further enhance the DT–SHM operability, which has been designed with multi-asset management in mind. Techniques such as deep learning for streaming data analysis, anomaly detection, unsupervised clustering for damage localisation, and predictive modelling for deterioration forecasting can increase automation and sensitivity to subtle structural changes, reducing reliance on manual calibration.

In parallel, the platform is inherently structured to operate across portfolios of assets, including different types of bridges, such as rail and road, suspension bridges, tunnels, and other infrastructure. It can evolve into a powerful asset management tool. By combining AI/ML-driven analytics with distributed digital twin ecosystems, asset managers will be able to conduct risk-based prioritisation, optimise maintenance planning, and allocate resources more efficiently at the network scale. This positions DT–SHM not only as a monitoring platform for individual structures but as a scalable, intelligent foundation for data-driven infrastructure management.

Scalability is another strategic focus. Although this framework was demonstrated on a heritage railway truss bridge, it is adaptable to diverse typologies, including suspension bridges and other large-scale assets. Scaling to such systems would require denser sensor networks to capture low-frequency dynamics, advanced modal analysis techniques, and edge computing to manage higher data volumes. These requirements highlight both the adaptability and the scalability challenges of extending the SHM–DT workflow to more complex infrastructure. Beyond bridges, the layered architecture can be applied to roadways, tunnels, and dams, and, when deployed across a network of assets, could form a distributed DT ecosystem for real-time portfolio management. Integration with smart city platforms would further link asset condition data with transport and urban systems, enhancing resilience planning at the network level.

The user interface and decision-support capabilities will also evolve. Incorporating augmented reality (AR) for on-site visual overlays of structural health data and virtual reality (VR) for immersive scenario simulation will make the DT platform more intuitive and accessible for engineers, inspectors, and non-technical stakeholders. These features can facilitate informed decision-making, accelerate training, and improve collaboration across disciplines.

Finally, achieving standardisation and interoperability will be critical for widespread adoption. Developing common protocols for data collection, model calibration, and validation will ensure reliability and comparability across different projects and jurisdictions. Coupled with open data standards, this will support collaborative research and foster innovation in DT–enabled SHM.

In summary, the future vision is a scalable, AI-enhanced, and user-centric SHM–DT ecosystem that continuously learns from data, adapts to evolving conditions, and operates seamlessly across infrastructure networks, transforming bridge monitoring from a periodic, reactive task into an autonomous, predictive, and integrated management process.

CRediT authorship contribution statement

Maryam Nasim: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Abbas Rajabifard:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Yiqun Chen:** Writing – review & editing, Software. **Bijan Samali:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

The authors declare the following financial and non-financial interests that could be perceived as influencing the submitted work:

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References

Ahmed, S.K., Haifa Afrin, Z., Shenoy, S., 2024. Free vibration response studies on bridges. IOP Conf. Ser. Earth Environ. Sci. 1387 (1). https://doi.org/10.1088/1755-1315/1387/1/012020.

Armijo, A., Zamora-Sánchez, D., 2024. Integration of railway bridge structural health monitoring into the internet of things with a digital twin: a case study. Sensors 24 (7), 2115. https://doi.org/10.3390/s24072115.

Bao, Y., Li, H., 2020. Machine learning paradigm for structural health monitoring. Struct. Health Monit. 20 (4), 1353–1372. https://doi.org/10.1177/1475921720972416.

Brenner, C., Unglaub, J., Thiele, K., 2023. A Reduced-Order Digital Twin for Structural Health Monitoring of Steel Bridges. https://doi.org/10.12783/shm2023/36798.

Casprini, E., Marini, A., Bartoli, G., Passoni, C., 2022. DEMSA protocol: deterioration effect modelling for structural assessment of RC buildings. Buildings 12 (5), 574. https://doi.org/10.3390/buildings12050574.

Chen, J., Reitz, J., Roßmann, J., Richstein, R., Schröder, K.-U., 2024. IoT-Based SHM using digital twins for interoperable and scalable decentralized smart sensing systems. Information 15 (3), 121. https://doi.org/10.3390/info15030121.

Chiang, K.-W., Li, Y.-H., El-Sheimy, N., Tsai, G.-J., 2017. Development of LiDAR-Based UAV system for environment reconstruction. IEEE Geosci. Remote Sensing Lett. 14. https://doi.org/10.1109/lgrs.2017.2736013.

Chopra, A.K., 2007. Dynamics of Structures. Pearson Education, India. https://doi.org/10.1002/9781118599792.

Dang, H.V., Tatipamula, M., Nguyen, H.X., 2022. Cloud-based digital twinning for structural health monitoring using deep learning. IEEE Trans. Ind. Inf. 18 (6), 3820–3830. https://doi.org/10.1109/TII.2021.3115119.

Das, S., Saha, P., Patro, S.K., 2016. Vibration-based damage detection techniques used for health monitoring of structures: a review. J. Civ. Struct. Health Monit. 6, 477–507. https://doi.org/10.1007/s13349-016-0168-5.

Escobar Villanueva, J.R., Iglesias Martinez, L., Perez Montiel, J.I., 2019. DEM generation from fixed-wing UAV imageing and LiDAR-derived ground control points for flood estimations. Sensors 19 (14), 3205. https://doi.org/10.3390/s19143205.

Fiandaca, D., Di Matteo, A., Patella, B., Moukri, N., Inguanta, R., Llort, D., Mulone, A., Mulone, A., Alsamahi, S., Pirrotta, A., 2022. An Integrated Approach for Structural Health Monitoring and Damage Detection of Bridges: An Experimental Assessment. Appl. Sci. 12 (24), 13018. https://doi.org/10.3390/app122413018.

Flah, M., Nunez, I., Ben Chaabene, W., Nehdi, M.L., 2021. Machine learning algorithms in civil structural health monitoring: a systematic review. Arch. Comput. Methods Eng. 28 (4), 2621–2643. https://doi.org/10.1007/s11831-020-09471-9.

García-Macías, E., Hernández-González, I.A., Ubertini, F., 2024. Incorporating Digital Twins and Artificial Intelligence for Next-Generation. In: Proceedings of SHM Software BT, Naples, Italy, pp. 435–447. https://doi.org/10.1007/978-3-031-61421-7_43.

Glaesssgen, E., Stargel, D., 2012. The digital twin paradigm for future NASA and US air force vehicles. https://doi.org/10.2514/6.2012-1818.

Gomez-Cabrera, A., Escamilla-Ambrosio, P.J., 2022. Review of machine-learning techniques applied to structural health monitoring systems for building and bridge structures. Appl. Sci. 12 (21), 10754. https://doi.org/10.3390/app122110754.

Grieves, M.W., 2023. In: Digital twins: past, present, and future. The Digital Twin. Springer, pp. 97–121. https://doi.org/10.1007/978-3-031-21343-4_4.

Hakimi, O., Alhawiti, A., Houshyar, A., Liu, H., Abudayyeh, O., Almatared, M., 2023. Data fusion for smart civil infrastructure management: a conceptual digital twin framework. Buildings 13 (11), 2725. https://doi.org/10.3390/buildings13112725.

Hao, J., Zhu, X., Li, J., 2024. Structural damage detection for spatial frame structures with semi-rigid joints using multiple set wireless measurements. J. Vib. Control 10775463241290044. https://doi.org/10.1177/10775463241290043.

Jeon, Y., Choi, G., Ahn, K., Lee, K.-H., Chang, S., 2025. Frequency-based finite element updating method for physics-based digital twin. Mathematics 13 (5), 738. https://doi.org/10.3390/math13050738.

Kioumarsi, M., Shabani, A., Plevris, V., 2019. Vulnerability Assessment of Cultural Heritage Structures. In: Proceedings of 53rd AIAA/ASME/ASCE/AHS/ASC Struclures, Structural Dynamics and Materials Conference. Hawaii, USA. https://doi.org/10.23967/eccomas.2022.294.

Kong, X., Hucks, R.G., 2023. Preserving our heritage: a photogrammetry-based digital twin framework for monitoring deteriorations of historic structures. Autom. ConStruct. 152, 104928. https://doi.org/10.1016/j.autcon.2023.104928.

Liu, X., Lieven, N.A.J., Escamilla-Ambrosio, P.J., 2009. Frequency response function shape-based methods for structural damage localisation. Mech. Syst. Signal Process. 23 (4), 1243–1259. https://doi.org/10.1016/j.ymssp.2008.10.002.

Liu, W., Lai, Z., Bacsa, K., Chatzi, E., 2023a. Neural extended Kalman filters for learning and predicting dynamics of structural systems. Struct. Health Monit. 23. https://doi.org/10.1177/14759217231179912.

Liu, Z., Meyendorf, N., Blasch, E., Yang, C., Tsukada, K., Liao, M., 2023b. Digital twin for predictive maintenance. Procedia CIRP 118, 306–311. In: https://doi.org/10.1016/j.procir.2023.06.053.

Loubet, G., Takacs, A., Gardner, E., De Luca, A., Udrea, F., Dragomirescu, D., 2019. LoRaWAN battery-free wireless sensors network designed for structural health monitoring in the construction domain. Sensors 19 (7), 1510. https://doi.org/10.3390/s19071510.

Mahmoodian, M., Shahrivar, F., Setunge, S., Mazaheri, S., 2022. Development of digital twin for intelligent maintenance of civil infrastructure. Sustainability 14 (14), 8664. https://doi.org/10.3390/su14148664.

Nasim, M., Setunge, S., 2022. Damage estimation of a concrete pier when exposed to extreme flood and debris loading. J. Mar. Sci. Eng. 10 (5), 710. https://doi.org/10.3390/jmse10050710.

Nasim, M., Setunge, S., 2024. A pioneering integration of structural health assessments and dynamic analyses: bridge pier responses to the impact of floating objects during extreme floods. J. Mar. Sci. Eng. 12 (4), 526. https://doi.org/10.3390/jmse12040526.

Ozbulut, O.E., Michael, R.J., Silwal, B., 2015. Dynamics of Civil Structures 2. https://doi.org/10.1007/978-3-319-15248-6.

Polonelli, T., Brunelli, D., Guermandi, M., Benini, L., 2018. An accurate low-cost Crackmeter with LoRaWAN communication and energy harvesting capability. In: Proceediings of IEEE International Conference on Emerging Technologies and Factory Automation. ETFA, Turin, Italy, pp. 671–676. https://doi.org/10.1109/ETFA.2018.8502592.

Qin, L.-F., Guo, C.-R., Ren, W.-X., 2023. A physics-data hybrid framework to develop Bridge digital twin model in structural health monitoring. Int. J. Struct. Stabil. Dynam. 23 (16), 2340037. https://doi.org/10.1142/S0219455423400370.

S. Australia, 2017. Part 7: bridge design – design loads (AS 5100.7-2017). Sydney, [Online]. Available: https://store.standards.org.au/product/as-5100-2-2017. .

Šašak, J., Gallay, M., Kaňuk, J., Hofierka, J., Minár, J., 2019. Combined use of terrestrial laser scanning and UAV photogrammetry in mapping alpine terrain. Remote Sens. 11 (18), 2154. https://doi.org/10.3390/rs11182154.

Geographical Area Network—structural health monitoring utility computing model. In: Tariq, H., Tahir, A., Touati, F., Al-Hitmi, M.A.E., Crescini, D., Ben Manouer, A. (Eds.), ISPRS Int. J. GeoInf. 8 (3), 154. https://doi.org/10.3390/ijgi8030154.

Torzoni, M., Tezzele, M., Mariani, S., Manzoni, A., Willcox, K.E., 2024. A digital twin framework for civil engineering structures. Comput. Methods Appl. Mech. Eng. 418, 116584. https://doi.org/10.1016/j.cma.2023.116584.

Vicente, R., Mendes Da Silva, J.A.R., Lagomarsino, S., Ferreira, T.M., Cattari, S., 2018. Cultural Heritage Monuments and Historical Buildings: Conservation Works and Structural Retrofitting. J. Build. Pathol. Rehabil. 9. https://doi.org/10.1007/978-981-10-5858-5_2.

Wang, T., Zhao, L., Jan, N., 2022. Virtual reality-based digital restoration methods and applications for Ancient buildings. J. Math. 2022. https://doi.org/10.1155/2022/2305463.

Wang, Q., Huang, B., Gao, Y., Jiao, C., 2025. Current status and prospects of digital twin approaches in structural health monitoring. Building 15 (7), 1021. https://doi.org/10.3390/buildings15071021.

Xu, Y., Zhang, J., 2022. UAV-based bridge geometric shape measurement using automatic bridge component detection and distributed multi-view reconstruction. Autom. ConStruct. 140, 104376. https://doi.org/10.1016/j.autcon.2022.104376.

Ye, X.W., Ni, Y.Q., Wong, K.Y., Ko, J.M., 2012. Statistical analysis of stress spectra for fatigue life assessment of steel bridges with structural health monitoring data. Eng. Struct. 45, 166–176. https://doi.org/10.1016/j.engstruct.2012.06.016.

Ye, C., Butler, L., Calka, B., Langurazov, M., 2019. A digital twin of bridges for structural. In: Prceedings of 12th International Workshop on Structural Health Monitoring 2019. California, USA. Stanford University. https://doi.org/10.12783/shm2019/32287.

Zhang, W., Mao, F., Qiu, H., 2011. Natural frequency of truss bridge in different support forms. In: Proceedings of Second International Conference on Mechanic Automation and Control Engineering, Hohhot, China, pp. 449–452. https://doi.org/10.1109/MACE.2011.5986956.

Zhou, C., Xiao, D., Hu, J., Yang, Y., Li, B., Hu, S., Demartino, C., Butala, M., 2021. An Example of Digital Twins for Bridge Monitoring and Maintenance: Preliminary Results. In: Proceedings of European Association on Quality Control of Bridges and Structures, Dublin, Ireland, pp. 1134–1143. https://doi.org/10.1007/978-3-030-91877-4_129.